

An astronomer, inventor, and mathematician, as well as a renowned writer and artist, Zhang Heng was one of ancient China's greatest thinkers. Among his scientific innovations was the first device for detecting earthquakes.

MILESTONES

CAREER CHANGE

Achieves fame for his writing by 108 CE; leaves his literary career to study science.

SERVES EMPEROR

Offered the role of chief astronomer in 115 CE; a highly prestigious post in the Chinese imperial court.

WRITES TREATISE

Publishes an innovative theory in 120 CE, proposing a spherical structure for the Universe and planets.

ADVANCES MATH

Improves on the Chinese formula for calculating π (pi), producing a figure of 3.162, in c.130 CE.

INVENTS SEISMOSCOPE

Builds a simple seismic device in 132 CE; it proves its accuracy by recording a distant earthquake.

Born during the Eastern Han Dynasty, whose rulers promoted science and technology, Zhang Heng achieved success as a literary scholar before turning his attention to astronomy and mathematics. The expertise he acquired in those fields led to appointments at the court of Emperor Shun, where he rose to become chief astronomer.

Discoveries and inventions

From his thorough observations of the night sky, Zhang concluded that the Moon and planets did not produce light, but reflected it from the Sun; he also believed that the Universe and planets were spherical. He mapped the stars and planets in a detailed catalog, identifying 124 constellations; 2,500 large stars; and 11,250 smaller ones.

A highly skilled engineer, Zhang constructed a water-powered armillary sphere: a model that depicted the night sky through a framework of rings, positioned to mirror the movements of celestial objects. He also designed a nonmagnetic compass and developed an instrument that recorded the direction in which an earthquake had occurred—this is regarded as the world's first seismoscope.

The basic principles of Zhang's earthquake-registering instrument subsequently featured in studies that led to the development of modern seismographs. His most notable contribution to the field of mathematics was his own calculation of the formula for π (pi); his figure would not be improved upon in China for another 200 years.

Zhang's seismoscope traced Earth tremors. On detection, a ball made a sound by dropping from one of the eight dragons' heads into the frog below it.

“The **sky** is like a hen's egg ... the **Earth** is like the **yolk** of the egg.”

Zhang Heng, 120 CE

